

YASHIKA SINGH

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Summary

AI/ML Engineer with around 4 years of experience building end-to-end AI products, from LLM-powered applications to ML models. Skilled in RAG, prompt engineering, and multimodal AI, with production experience delivering projects at Product Manager Accelerator under Dr. Nancy Li and as Lead AI Engineer at Columbia University's Summer Startup Program. Selected for the prestigious MOVE Program by Handshake, recognizing top Master's students for leadership and innovation in technology.

Work Experience

Model Validation Expert Fellow, Handshake, San Francisco, CA (Remote Part-Time) Sep 2025 - Present

- Collaborating with leading researchers to refine the capabilities of Large Language Models (LLMs) by designing diverse prompts, contributing to testing, evaluation, and contextual improvements in AI systems.
- Inducted as a Star Fellow, a distinction reserved for top-performing fellows recognized for creating the highest-quality prompts and contributions to AI research and evaluation.

Lead AI Engineer, Columbia University Summer Startup Program, New York July 2025 - Aug 2025

- Collaborated with Columbia University student founders on a fintech startup initiative, providing AI leadership from concept to MVP and achieved a successful launch within 8 weeks, including the end-to-end design of RAG pipelines, LLM integrations, and product-facing dashboards.
- Architected and implemented Retrieval-Augmented Generation (RAG) workflows in Python/Django with vector search (e.g., FAISS/pgvector) and PostgreSQL, owning both backend services and frontend user experience.
- Wrote and maintained production-grade Python services (FastAPI/Django) with unit tests and code reviews, ensuring reliability as traffic and complexity increased.
- Delivered client-facing dashboards, technical documentation, and an implementation roadmap that guided post-program development and investor demos.

AI/ML Engineer Intern, Product Manager Accelerator, Boston, MA (Remote) Jan 2025 - Present

- Designed and deployed AI-driven interview simulation workflows using agentic AI for Product Manager Accelerator's proprietary platform, directly under the leadership of founder Dr. Nancy Li.
- Leveraged RAGs and LLMs (OpenAI, Claude) to generate context-aware product management questions and applied NLP techniques and prompt engineering to build dynamic follow-up logic and structured candidate evaluation rubrics, improving depth and consistency of interviewer feedback.
- Improved real-time response latency by ~70% by redesigning the model pipeline (caching, batching, retrieval tuning) and shipped it as a production service in Python (APIs, tests, logging), participating in code reviews and on-call debugging.

Data Scientist, Axtia, Noida, India Aug 2021 - Dec 2023

- Optimized Python scripts and SQL queries for pharma-commercial datasets, reducing end-to-end data processing time on key workflows.
- Developed and maintained ETL pipelines using Matillion and SQL to integrate multi-source datasets into analytical models for US life-sciences clients.
- Automated complex data workflows using DAGs in Apache Airflow and Azure Data Factory processing 10M+ records per daily run with monitored SLAs, reducing manual intervention by ~40% and improving data reliability in recurring production loads.
- Integrated MLOps best practices with CI/CD pipelines to automate model training, deployment and monitoring on Databricks, ensuring scalability and reliability.
- Created interactive dashboards and data visualizations using Power BI and Tableau to deliver insights and support decision-making for internal stakeholders.

Projects

Drowsiness Detection for Drivers: [GitHub Link]

Built a CNN-based, real-time system using TensorFlow to detect driver fatigue and improve road safety with validation accuracy around 95%.

Detection of Cardiomegaly in Dogs: [GitHub Link]

Created a deep learning model using custom CNN for X-ray analysis and automating veterinary diagnostics with an accuracy of 76.75%, surpassing the benchmark set by VGG16 (75%).

Education

Yeshiva University, Katz School of Science and Health, New York, NY

Master of Science in Artificial Intelligence, GPA: 3.8/4.0

May 2025

Galgotias College of Engineering and Technology, Greater Noida, India

Bachelor of Technology in Computer Science, CGPA: 3.5/4.0

Aug 2021

Technical Skills

Programming Languages: Python, C, C++, JavaScript, SQL

Data & Backend: Apache Airflow (DAGs), Azure Data Factory, Databricks, Spark/PySpark, Django, Vector Databases, Extract-Transform-Load (ETL), MySQL, PostgreSQL, AWS, Google Cloud Platform (GCP), Salesforce Data Cloud

ML & Generative AI: Natural Language Processing (NLP), Generative AI, Large Language Models (LLM), Prompt Engineering, Retrieval-Augmented Generation (RAG), Hugging Face, LangGraph, LangChain, Machine Learning, Deep Learning, Data Science, Convolutional Neural Networks (CNN), Computer Vision, Automatic Speech Recognition (ASR), Whisper, PyTorch, TensorFlow, Scikit-learn

Analytics, Visualization & Tools: Data Analysis, Exploratory Data Analysis (EDA), Pandas, Matplotlib, Streamlit, Tableau, Power BI, Jupyter, Google Colab, Anaconda, VS Code, Git, React, NodeJS, HTML5, CSS